deeply immersed in black-hole research) thought yes. I thought no. In November 1971, we made a bet.

The tools did not yet exist for resolving the bet. What kinds of tools were needed? Since the pulsations would begin weak and only gradually grow strong (if they grew at all), they could be regarded as small "perturbations" of the hole's spacetime curvature—just as the vibrations of a ringing wine glass are small perturbations of the glass's shape. This meant that the hole's pulsations could be analyzed using the perturbation methods whose spirit was described in Box 7.1 above. However, the specific perturbation methods which Price, Press, Vish-veshwara, Chandrasekhar, and others were using in the autumn of 1971 would work only for perturbations of nonspinning, or very slowly spinning, black holes. What they needed were entirely new perturbation methods, methods for perturbations of rapidly spinning holes.

The effort to devise such perturbation methods became a hot topic in 1971 and 1972. My students, Misner's students, Wheeler's students, and Chandrasekhar with his student John Friedman all worked on it, as did others. The competition was stiff. The winner was Saul Teu-kolsky, a student of mine from South Africa.

Teukolsky recalls vividly the scene when the equations of his method fell into place. "Sometimes when you play with mathematics, your mind starts picking out patterns," he says. "I was sitting at the kitchen table in our apartment in Pasadena one May evening in 1972, playing with the mathematics; and my wife Roz was making crepes in a Teflon pan, which was supposed not to stick. The crepes kept sticking. Everytime she poured the batter in she would bang the pan on the countertop. She was cursing and banging, and 1 was yelling at her to be quiet because 1 was getting excited; the mathematical terms were starting to cancel each other in my formulas. Everything was canceling! The equations were falling into place! As I sat there staring at my amazingly simple equations, I was filled with this feeling of how dumb I had been; I could have done it six months earlier; all I had to do was collect the right terms together."